

COMPLETE MULTI-RESTAURANT MANAGEMENT SYSTEM

Commercial SaaS Product Specification

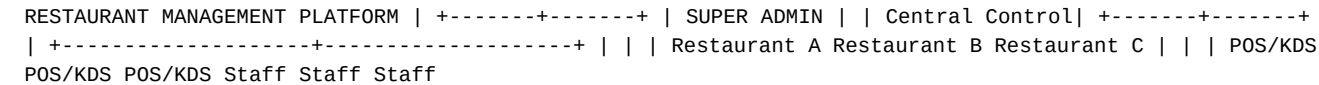
A complete specification for a restaurant management platform that can be sold to multiple restaurants, with isolated restaurant data and centralized Super Admin control.

1. Product Overview

The system is a **multi-tenant restaurant management SaaS platform**. One platform serves many independent restaurants. Each restaurant has its own users, branches, menu, tables, orders, customers, inventory, employees, reports and settings. A Super Admin has centralized visibility and management across all restaurants.

The platform should be designed for commercial resale from the beginning. Restaurant data must be isolated by tenant, while the platform owner can manage restaurants, subscriptions, plans, platform users and global analytics.

2. High-Level Architecture



Recommended application structure:

- **Backend:** Laravel REST API
- **Web application:** React or Vue
- **Mobile application:** React Native where mobile workflows are required
- **Database:** PostgreSQL
- **Cache/queues:** Redis
- **Real-time:** WebSockets for kitchen, POS and order status updates
- **Storage:** S3-compatible object storage for menu images, receipts and documents
- **Deployment:** Docker + Nginx with separate production and staging environments

3. Multi-Tenant Model

Every restaurant is a tenant. Restaurant-owned records should contain a restaurant identifier and, where relevant, a branch identifier. Application authorization must enforce tenant scope on every request.

restaurants	id		name	1		Restaurant A	2		Restaurant B	3		Restaurant C	orders	id		restaurant_id														
branch_id		total	1		1		1		500	2		1		1		800	3		2		3		1200	4		3		4		700

Important: Restaurant A must never be able to query or view Restaurant B's records. Super Admin is the only role with platform-wide visibility, subject to permissions.

4. Super Admin Portal

- Dashboard with total restaurants, active restaurants, branches, users, orders, sales and subscription status.
- Restaurant creation, activation, suspension, archiving and management.
- View restaurant branches, users, orders, sales, inventory and reports.
- Subscription plans, pricing, trials, renewals and feature limits.
- Global order and sales analytics with restaurant, branch and date filters.
- Platform user and permission management.
- Audit logs showing important administrative and financial actions.
- System-wide configuration, notifications and platform settings.

5. Restaurant Owner / Manager Portal

- Restaurant dashboard
- POS
- Orders
- Kitchen Display System
- Tables and floor management
- Reservations
- Menu and product management
- Customers / CRM
- Inventory
- Recipes / food costing
- Suppliers and purchasing
- Expenses
- Employees and roles
- Attendance and shifts
- Reports
- Restaurant and branch settings

6. POS System

The POS is the core operational module and should support:

- Dine-in, takeaway and delivery orders
- Table selection and table transfer
- Quick sale
- Hold and resume orders
- Split bills and merge bills
- Discounts, coupons and promotions
- Taxes and service charges
- Tips
- Cash, card, QR and other configured payment methods

- Multiple payment methods on one order
- Refunds and cancellations with permission control
- Receipt printing and reprinting
- Customer assignment
- Order notes and item-level notes

7. Table & Floor Management

- Create multiple floors or dining areas.
- Create tables with capacity and table numbers.
- Visual table status: Available, Occupied, Reserved, Cleaning and Out of Service.
- Move or transfer an active table.
- Merge tables.
- Assign orders to tables.
- Display active order total and elapsed time.

8. Menu Management

- Categories and subcategories
- Products / menu items
- Product images
- Descriptions
- Selling price and optional cost price
- SKU / barcode
- Tax configuration
- Availability / sold-out control
- Preparation time
- Featured / recommended items
- Multiple sizes and variants
- Modifiers and add-ons
- Item-specific instructions

Example modifier structure:

Pizza → Size → Small / Medium / Large; Toppings → Cheese / Mushroom / Chicken

9. Kitchen Display System (KDS)

Orders should move through a controlled kitchen workflow:

NEW ORDER → ACCEPTED → PREPARING → READY → SERVED / COMPLETED

- Kitchen order queue
- Order number and table
- Items and quantities
- Modifiers and preparation notes

- Elapsed preparation time
- Priority orders
- Station-based routing such as Hot Kitchen, Grill, Bar and Dessert
- Real-time status updates back to POS

10. QR Table Ordering

- Unique QR code for each table
- Customer scans without installing an application
- Digital menu
- Item customization
- Cart and order submission
- Table-aware ordering
- Customer notes
- Order status
- Optional online payment

11. Online Ordering

- Restaurant-specific public ordering page
- Pickup and delivery
- Menu browsing and item customization
- Customer account or guest checkout
- Delivery address
- Order confirmation
- Payment integration
- Order tracking
- Order history

A tenant can use a branded subdomain such as **restaurant-name.yourplatform.com**.

12. Reservations

- Reservation calendar
- Customer name and contact
- Guest count
- Date and time
- Table assignment
- Special requests
- Pending, Confirmed, Seated, Completed, Cancelled and No-Show statuses
- Reservation history

13. Customer / CRM

- Customer profile
- Phone and email
- Address
- Order history
- Total spending
- Visit count
- Favorite products
- Customer notes
- Loyalty points
- Membership level
- Promotional offers

14. Loyalty & Promotions

- Configurable points rules
- Rewards
- Membership tiers
- Coupons
- Discount campaigns
- Birthday promotions
- First-order promotions
- Restaurant-specific promotional rules

15. Inventory Management

Inventory must support both simple stock items and ingredient-level stock tracking.

- Ingredients and raw materials
- Packaging materials
- Stock units and conversion units
- Stock in / stock out
- Stock adjustment
- Stock transfer between branches
- Wastage
- Damaged and expired stock
- Stock count
- Minimum stock and low-stock alerts
- Stock movement history

16. Recipe / BOM & Food Costing

Recipes connect menu products to ingredients so sales can automatically consume inventory.

Chicken Burger Bread 1 pc Chicken 150 g Cheese 1 pc Lettuce 30 g Sauce 20 g

- Ingredient quantities per product
- Ingredient cost calculation
- Recipe versioning
- Automatic stock deduction from completed sales
- Food cost percentage
- Gross margin and estimated profit per product

17. Purchasing & Suppliers

- Supplier management
- Purchase requests
- Purchase orders
- Goods receiving
- Purchase invoices
- Purchase returns
- Supplier balances
- Supplier payments
- Purchase history

Purchase Request → Purchase Order → Receive Goods → Inventory → Supplier Invoice → Payment

18. Expenses

- Rent
- Electricity
- Water
- Internet
- Salary
- Transportation
- Maintenance
- Marketing
- Other operating expenses
- Expense categories
- Approval workflow where required
- Expense reports by date, branch and category

19. Employee Management

- Employee profiles
- Restaurant and branch assignment
- Role assignment
- Permissions
- Cashier, waiter, kitchen, manager, accountant and other configurable roles

- Employee status
- Shift assignment
- Attendance

20. Attendance & Shifts

- Clock in / clock out
- Break tracking
- Shift schedules
- Late arrival
- Absence
- Overtime
- Attendance reports
- Branch-level staff scheduling

21. Role-Based Access Control

Use permissions instead of hard-coded access rules.

- super_admin
- restaurant_owner
- restaurant_manager
- cashier
- waiter
- kitchen_staff
- accountant
- delivery_staff
- custom roles

Example permissions: orders.view, orders.create, orders.cancel, orders.refund, products.manage, inventory.view, inventory.adjust, reports.view, financial_reports.view, employees.manage.

22. Reports & Analytics

Sales Reports

- Daily, weekly, monthly and yearly sales
- Branch comparison
- Hourly sales
- Product sales
- Category sales
- Order type sales
- Payment method summary

Financial Reports

- Revenue

- Discounts
- Taxes
- Service charges
- Expenses
- Food cost
- Gross profit / margin
- Payment reconciliation

Inventory Reports

- Current stock
- Stock movement
- Wastage
- Purchases
- Low stock
- Food cost

Customer Reports

- New customers
- Returning customers
- Top customers
- Customer spending
- Visit frequency

Employee Reports

- Sales by employee
- Orders by employee
- Attendance
- Shift performance

23. Multi-Branch Management

A restaurant company may operate multiple branches. Branches share the tenant but have separate operational data where appropriate.

Restaurant Company ■■■ Branch 1 ■■■ Branch 2 ■■■ Branch 3 ■■■ Branch 4

- Branch-specific POS
- Branch-specific tables
- Branch inventory
- Branch employees
- Branch expenses
- Branch sales reports
- All-branch consolidated reports

24. Notifications

- New order
- New online order
- Kitchen status
- Low stock
- Reservation
- Payment confirmation
- Refund
- Subscription expiring
- Subscription expired
- Staff notifications

Delivery channels can include in-app notifications, email, SMS, push notifications and integrations appropriate to the target market.

25. Audit Logs

Important actions should be recorded for security and accountability.

03 Sep 2026 09:31 User: Manager Action: Updated product Product: Chicken Burger Old Price: 450 New Price: 500

- User
- Restaurant
- Branch
- Action
- Entity and record ID
- Old and new values where appropriate
- Timestamp
- IP / device metadata where legally and operationally appropriate

26. Subscription / SaaS Management

Because the platform will be sold to multiple restaurants, subscription management should be a first-class module.

Plan	Example limits	Example features
Starter	1 branch / 5 users	POS, Menu, Orders, Basic Reports
Professional	5 branches / 25 users	KDS, Inventory, CRM, Advanced Reports, Online Ordering
Enterprise	Custom	Unlimited branches/users, API, advanced analytics, dedicated support

- Free trial
- Monthly / yearly billing
- Subscription activation and expiration
- Plan upgrades and downgrades
- Feature limits
- Payment status

- Suspension after non-payment
- Invoices and subscription history

27. Restaurant Onboarding

Super Admin → Add Restaurant → Restaurant Information → Owner Account → Plan → Branch → Initial Settings → Activate

- Restaurant name and branding
- Owner account
- Address and contact information
- Tax configuration
- Currency
- Time zone
- Business hours
- Initial branch
- Subscription plan
- Default roles and permissions

28. Database Modules

Core database tables should be planned around the following domains:

- tenants / restaurants
- branches
- users
- roles
- permissions
- role_permissions
- user_roles
- categories
- products
- product_variants
- modifiers
- modifier_options
- tables
- floors
- orders
- order_items
- payments
- refunds
- kitchen_orders / kitchen_items
- reservations

- customers
- loyalty_accounts / loyalty_transactions
- inventory_items
- inventory_transactions
- recipes / recipe_items
- suppliers
- purchase_orders / purchase_items
- goods_receipts
- expenses
- employees
- shifts
- attendance
- subscriptions
- plans
- plan_features
- audit_logs
- notifications

29. Core Order Workflow

CUSTOMER / WAITER / CASHIER ↓ CREATE ORDER ↓ PAYMENT (if required) ↓ KITCHEN ↓ PREPARING ↓ READY ↓ SERVED / COMPLETED ↓ INVENTORY + REPORTING

30. Security Requirements

- Tenant-level authorization on every restaurant-owned resource.
- Role-based permissions.
- Secure password hashing and authentication.
- Token/session expiration and revocation.
- Server-side validation for every API request.
- Protection against SQL injection, XSS, CSRF where applicable and mass assignment.
- Rate limiting for sensitive endpoints.
- Audit logging for administrative and financial actions.
- Encrypted transport using HTTPS.
- Regular database backups and tested restoration procedures.
- Do not expose one restaurant's identifiers or data to another tenant.

31. API Design

Use versioned APIs, for example */api/v1/*, and group endpoints by module.

- Authentication API
- Restaurant / branch API

- User / role / permission API
- Menu API
- Table API
- POS / order API
- Payment API
- Kitchen API
- Reservation API
- Customer API
- Inventory API
- Recipe API
- Purchase / supplier API
- Expense API
- Employee API
- Report API
- Subscription API
- Super Admin API

32. Recommended Development Phases

Phase 1 — Core

- Authentication
- Restaurant and branch management
- Users and roles
- Menu
- Tables
- POS
- Orders
- Payments
- Receipts
- Dashboard

Phase 2 — Operations

- KDS
- Reservations
- QR ordering
- Online ordering
- Customers
- Inventory
- Recipes
- Purchasing

- Suppliers
- Expenses

Phase 3 — Business

- Advanced reports
- Loyalty
- Promotions
- Employee attendance
- Multi-branch analytics
- Notifications

Phase 4 — SaaS

- Subscriptions
- Plans
- Trials
- Feature limits
- Billing
- Super Admin analytics
- Tenant management
- Audit logs

33. Recommended Final Product Structure

Restaurant SaaS ■ ■ ■ ■ Super Admin ■ ■ ■ ■ Dashboard ■ ■ ■ ■ Restaurants ■ ■ ■ ■ Subscriptions ■ ■ ■ ■ Plans ■ ■ ■ ■ Users ■ ■ ■ ■ Global Orders ■ ■ ■ ■ Global Sales ■ ■ ■ ■ Global Reports ■ ■ ■ ■ Audit Logs ■ ■ ■ ■ System Settings ■ ■ ■ ■ Restaurant Owner / Manager ■ ■ ■ ■ Dashboard ■ ■ ■ ■ POS ■ ■ ■ ■ Orders ■ ■ ■ ■ Kitchen ■ ■ ■ ■ Tables ■ ■ ■ ■ Reservations ■ ■ ■ ■ Menu ■ ■ ■ ■ Customers ■ ■ ■ ■ Inventory ■ ■ ■ ■ Recipes ■ ■ ■ ■ Suppliers ■ ■ ■ ■ Purchases ■ ■ ■ ■ Expenses ■ ■ ■ ■ Employees ■ ■ ■ ■ Reports ■ ■ ■ ■ Settings ■ ■ ■ ■ Customer ■ ■ ■ ■ Menu ■ ■ ■ ■ QR Order ■ ■ ■ ■ Online Order ■ ■ ■ ■ Reservation ■ ■ ■ ■ Order Tracking ■ ■ ■ ■ Kitchen ■ ■ ■ ■ Incoming Orders ■ ■ ■ ■ Preparing ■ ■ ■ ■ Ready ■ ■ ■ ■ Completed

34. Commercial Goal

The final product should behave as a reusable SaaS platform rather than a one-off restaurant application. A new restaurant should be onboarded without deploying a new application. The Super Admin should be able to activate a restaurant, assign a subscription plan, create its owner account and monitor the restaurant from the same central platform.

Key principle: Build the tenant architecture, authorization, database relationships and subscription model correctly at the beginning. These are the foundation for safely selling the same system to many restaurants.